

Tutorial No. 7

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Problem Statement - Analyze different types of software maintenance and identify the appropriate maintenance strategy for a real-world software failure.

Solution -

Introduction

→ Software maintenance is a crucial phase in the SDLC. After deployment, software systems require continuous updates to fix errors, improve performance, adapt to ~~envi~~ environmental changes, and enhance features. According to IEEE, software maintenance involves modification of a software product after delivery to correct faults, improve performance or adapt to a changed environment.

2. Types of Software Maintenance

→ Software maintenance is generally classified into four major types:

2.1 Corrective Maintenance

- Fixes bugs and defects discovered after deployment.
- Triggered by user-reported errors or system failures.
- Example: Fixing a login authentication crash.

1. Adaptive Maintenance
Modified software to remain compatible with changing environments
Required when OS, hardware, or third-party APIs change.
Example: Updating software to support a new version of windows or android.

2. Perfective Maintenance
Enhanced system performance or adds new features based on user feedback.
Improves usability and efficiency
Example: Adding a new report generation module

3. Preventive Maintenance
Improved maintainability and reliability to prevent future issues.
Includes code refactoring and optimization
Example: Removing deprecated functions to reduce future risks.

4. Real-World Software Failure Scenarios
Example: Payment gateway failure in an e-commerce system.

→ A large e-commerce platform experienced frequent payment failures during peak traffic. Users were unable to complete transactions, causing financial loss and customer dissatisfaction.

Observed issues:

- Server crash under heavy load.
- Timeout errors during transaction processing.
- Incompatibility with updated bank API.

Analysis of the failure.

- The system failure involved multiple root causes:
 - System crash due to overload → Performance issue
 - Timeout errors → Possible bug in transaction handling.
 - API incompatibility → Environmental change.
- This indicates that more than one maintenance type is required

Appropriate Maintenance Strategy.

- 1. Corrective Maintenance
 - Fix timeout bugs.
 - Patch transaction handling logic.
- 2. Adaptive Maintenance
 - Update integration to match the new banking API version.
 - Ensure compatibility with updated security protocols.
- 3. Perfective Maintenance
 - Optimize database queries.
 - Implement load balancing and caching
 - Improve server scalability
- 4. Preventive Maintenance
 - Refactor payment module code
 - Perform stress testing

Implement automated monitoring tools.

Recommended Maintenance Strategy

The most appropriate strategy in this case is a Hybrid Maintenance Strategy combining:
Immediate Corrective Maintenance to restore service

- Adaptive Maintenance to ensure compatibility
 - Perfective Maintenance to improve performance
 - Preventive Maintenance to avoid recurrence
- A structured approach like the DevOps maintenance model (continuous monitoring + CI/CD updates) is recommended for ~~the~~ long-term reliability.

Conclusion

Software maintenance is not limited to fixing bugs; it ensures long-term system stability, scalability, and adaptability. In real-world failures ~~at~~ such payment gateway crashes, a combination of corrective, adaptive, perfective, and preventive maintenance strategies is essential. Proper analysis and structured maintenance planning reduce downtime, financial loss, and ~~so~~ customer dissatisfaction.

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